

Statement of Work
for
Manikin Replacement Parts
at the
704th Test Group/846th Test Squadron

26 January 2026

1.0 PURPOSE

The 846 TS operates the Holloman High Speed Test Track (HHSTT) and performs rocket sled testing of USAF aircraft emergency escape systems using mechanical surrogates known as biofidelic instrumented anthropomorphic test devices (ATDs, or “manikins”) in the place of live aircrew. The two main types of manikins used are Joint Primary Aircraft Training System (JPATS) Case-1 Reduced Weight, also known as the Lightest Occupant in Service (LOIS) weighing 103 lb and the JPATS Case-6, also known as the Large Anthropomorphic Research Dummy (LARD) weighing 245 lb. These are custom-made manikins which have specific anthropometry and mass properties of individual body parts and limb segments. Because the LOIS and LARD ATDs are custom-made for the Department of War (DoW), most replacement parts are not commercially available off-the-shelf.

The DoW requires using ATDs with sled testing for the qualification of egress systems and other aircrew equipment prior to fielding, in accordance with the Crew Systems Engineering Bulletin EZFC-CSB-16-001 (Revision of MIL-HDBK-516C Airworthiness Certification Criteria, AFLCMC/Crew Systems Engineering, 28 November 2016).

As expected, these HHSTT manikins have parts that break during the execution of these particularly severe escape system tests. This document explains the requirements for purchase of unique replacement manikin mechanical parts designed and manufactured by Humanetics Innovative Solutions, Inc. (“Humanetics”) that will return two particular manikins (LOIS-9 and LARD-23) back to service after they were damaged in egress testing (see tables-1 and -2 for required parts listing description and quantities). Part numbers in tables-1 and -2 are proprietary Humanetics part numbers for LOIS and LARD manikins, of the same type contained in the manikin assemblies recently delivered under contract# FA860624CB002, ATD Solution.

1.1 LOIS Part Numbers, Descriptions and Quantities:

Table 1: Required LOIS Parts

Humanetics Item Part Number	Description	Unit	Quantity
880105-439-FT	FEMUR ASSEMBLY, MACHINED, RIGHT	EA	1
200-4901-2-PR	FEMUR COVER, RIGHT, LOIS	EA	1
5001054-FT	SCREW LHCS M4 X 0.7 X 12	EA	1
240-6110-H	UPPER ARM MOLDED ASSEMBLY	EA	1
240-6510-H	LOWER ARM ASSEMBLY, MOLDED	EA	1
240-6150	UPPER ARM LOWER PART	EA	1
880105-344-DN	RUBBER STOP ASSEMBLY	EA	1
880105-717-DN	LOWER ARM STOP ASSEMBLY	EA	2

1.2 LARD Part Numbers, Descriptions and Quantities:

Table 2: Required LARD Parts

Item Part Number	Description	Unit	Quantity
78051-198-DN	STOP ASSEMBLY LOWER ARM	EA	4
190-6510-H	LOWER ARM MOLDED ASSY	EA	1
890520-0611-FT	OUTBOARD ASSY,STRUCTURAL REPLACEMENT KNEE	EA	1
190-102	CAP SKIN JSF-LM110	EA	1
79051-34-H	ROTATION STOP COVER	EA	1

2.0 SCOPE

The Contractor shall deliver the manikin parts listed in table-1 and table-2 in section 1.0 above. These parts will meet the following minimum requirements: .

2.1 Mechanical Requirements:

- 2.1.1 Parts will have the proper dimensions, weight, center of gravity, hardness, and biofidelity (if applicable) as defined by 711 HPW/RH, AFLCMC/ROU, AFLCMC/EZFC and 704 TG/846 TS and implemented for production by Humanetics via their manufacturing process specifications for JPATS-based LOIS and LARD entire manikin assemblies (see contract# FA860624CB002).
- 2.1.2 Parts will have the same characteristics (enumerated in 2.1.1), which were previously characterized and verified by 711 HPW/RH. .

2.2 Documentation and Support:

- 2.2.1 Parts illustrations and nomenclature will be included in the exploded assembly diagrams contained with the LOIS and LARD product manuals.
- 2.2.2 Parts handling, installation procedure and maintenance details (if applicable) as described in the LOIS and LARD product manuals.

3.0 MILESTONES

The contractor shall deliver the prescribed manikin parts within 20 to 24 weeks (or better) after receiving the order.

4.0 DELIVERABLES

The contractor shall deliver all hardware and documentation to 846 TS/TGTMI, Attn: Timothy Gros, at 1521 Test Track Road, building 1179, Holloman Air Force Base, New Mexico 88330-7850.